

BFS Traversal

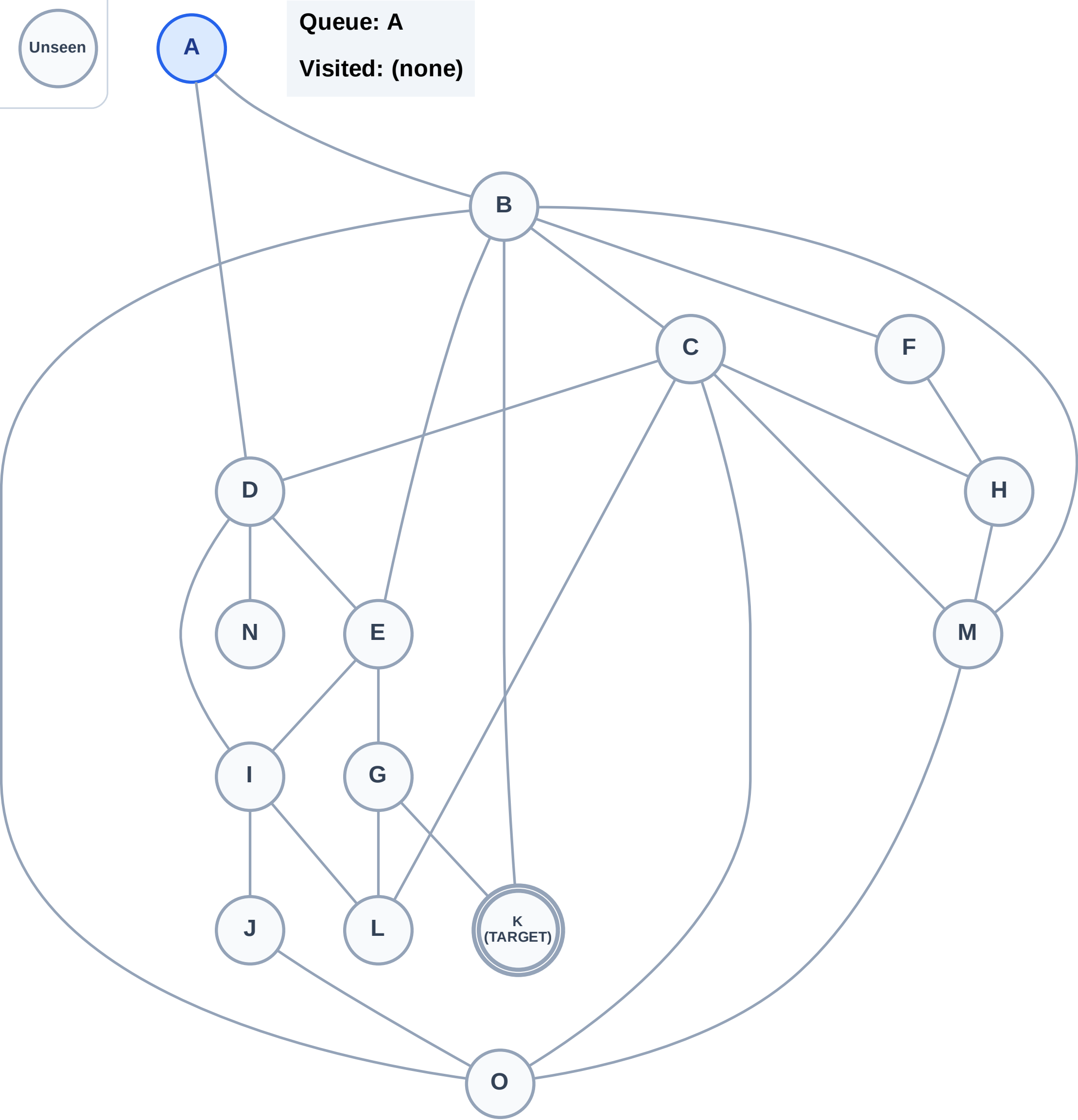
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Complete

Processing

Discovered

Unseen



BFS Traversal

Step 2: Process node A

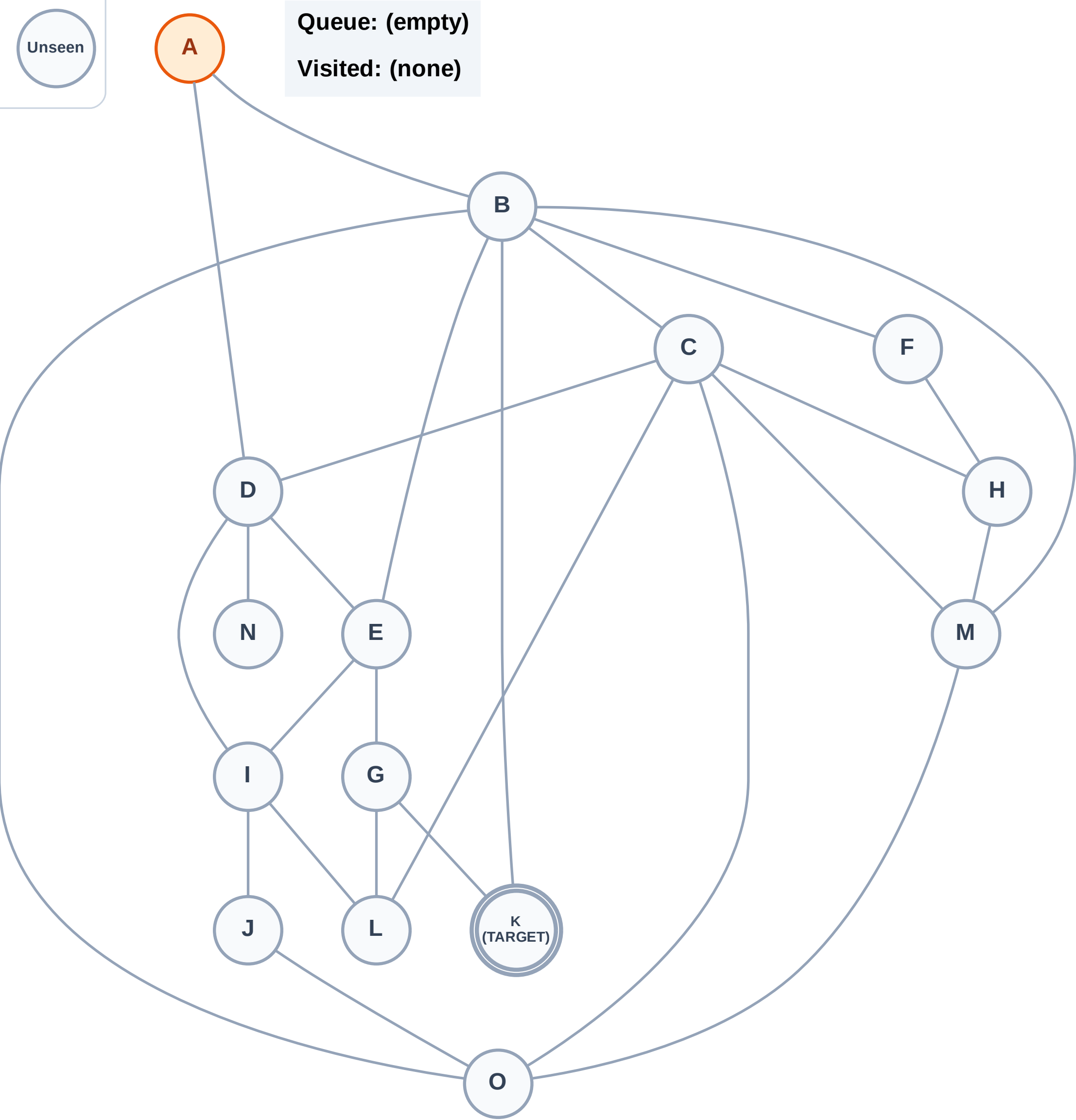
Legend

Complete

Processing

Discovered

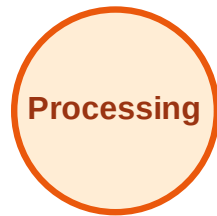
Unseen



BFS Traversal

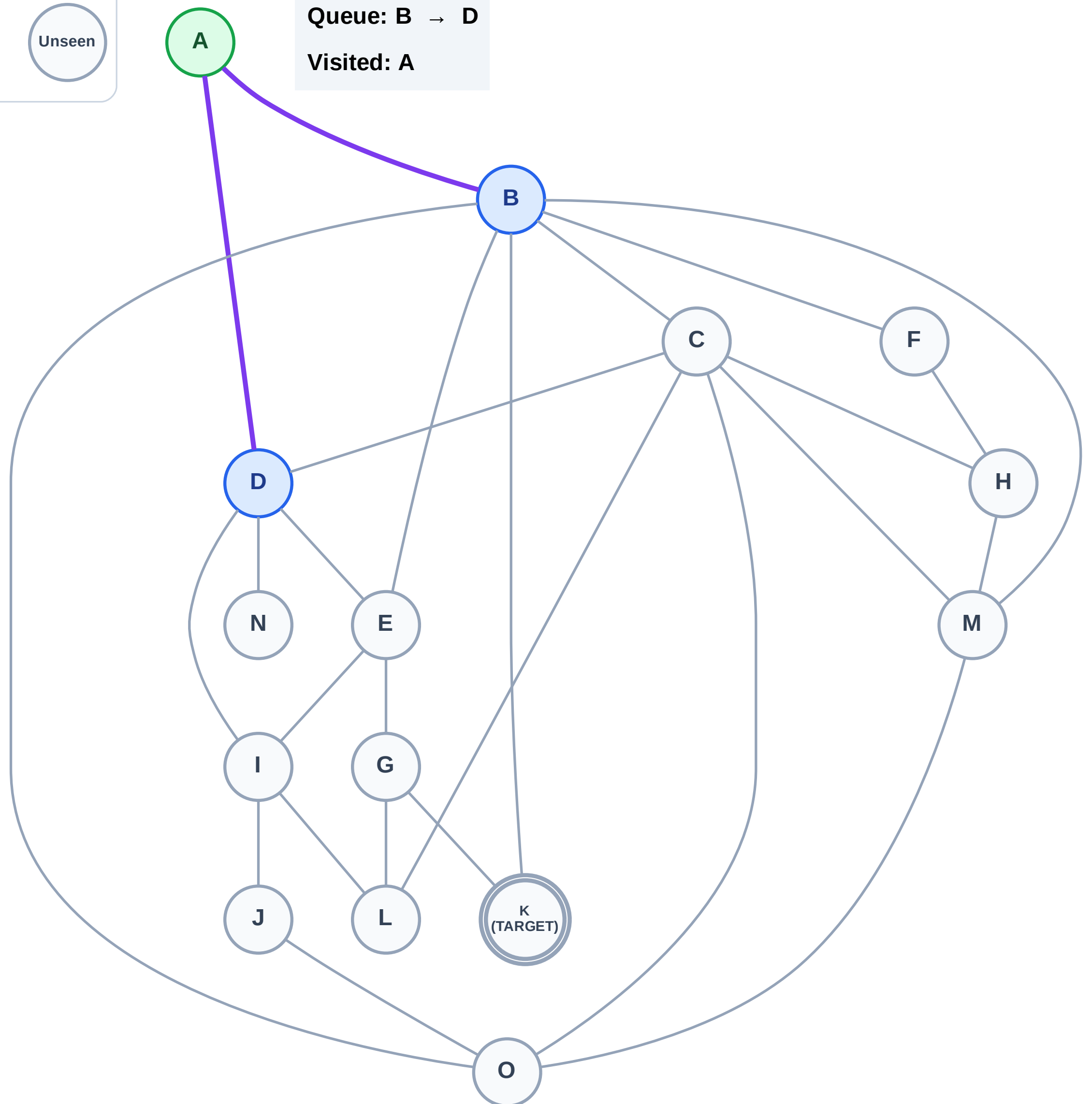
Step 3: Discover B, D from A

Legend



Queue: B → D

Visited: A



BFS Traversal

Step 4: Process node B

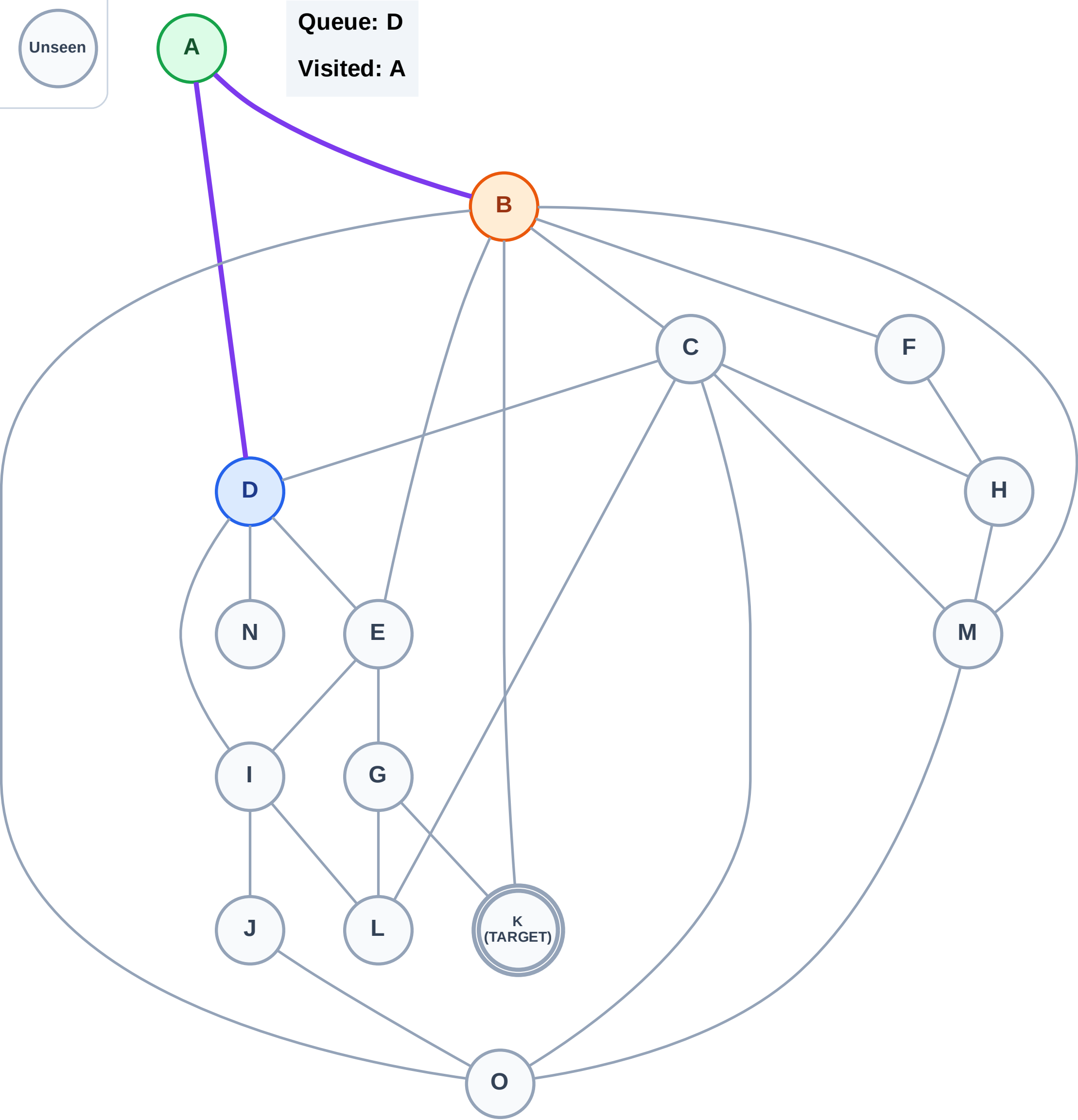
Legend

Complete

Processing

Discovered

Unseen



BFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

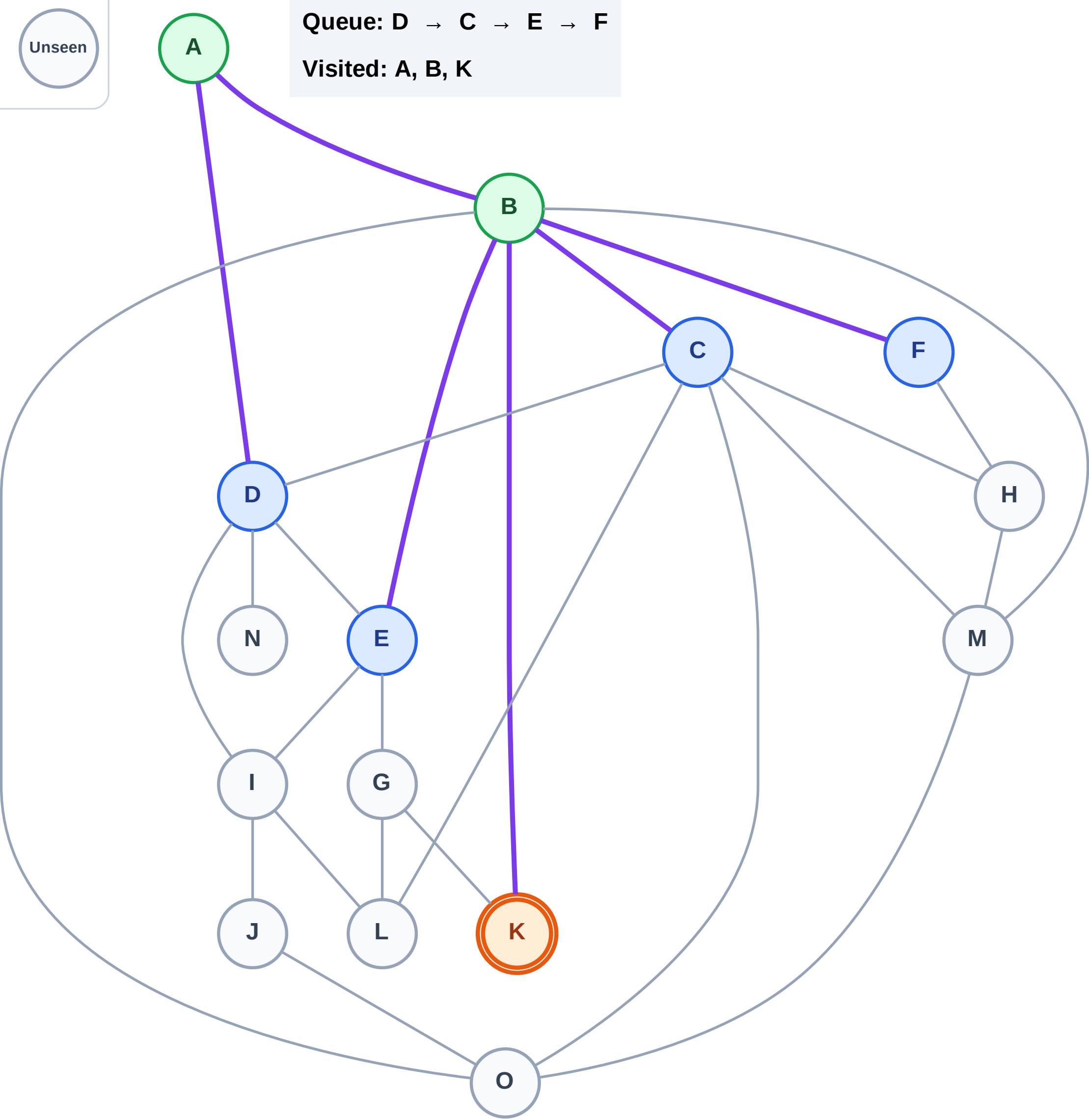
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

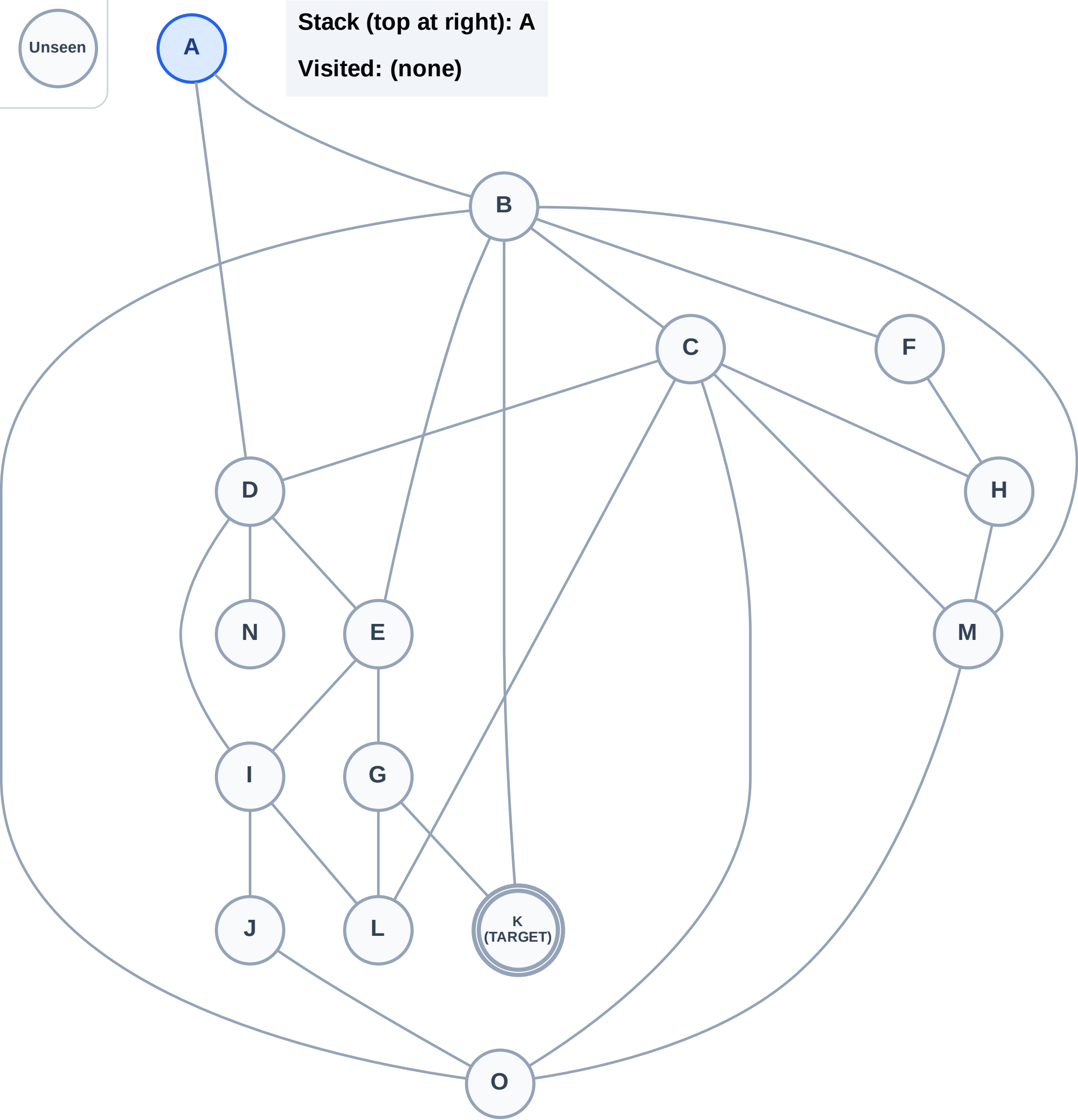
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 2: Process node A

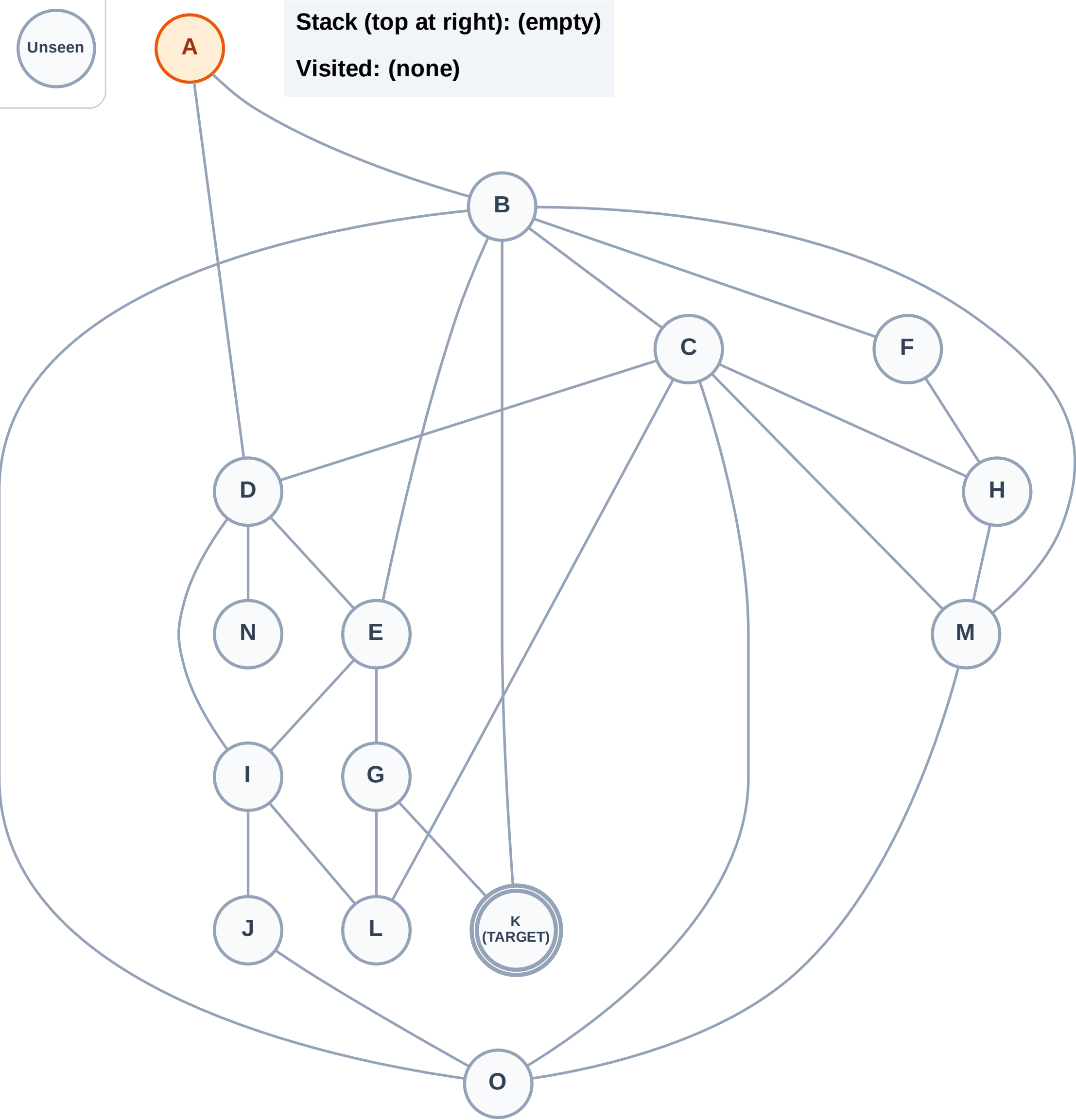
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

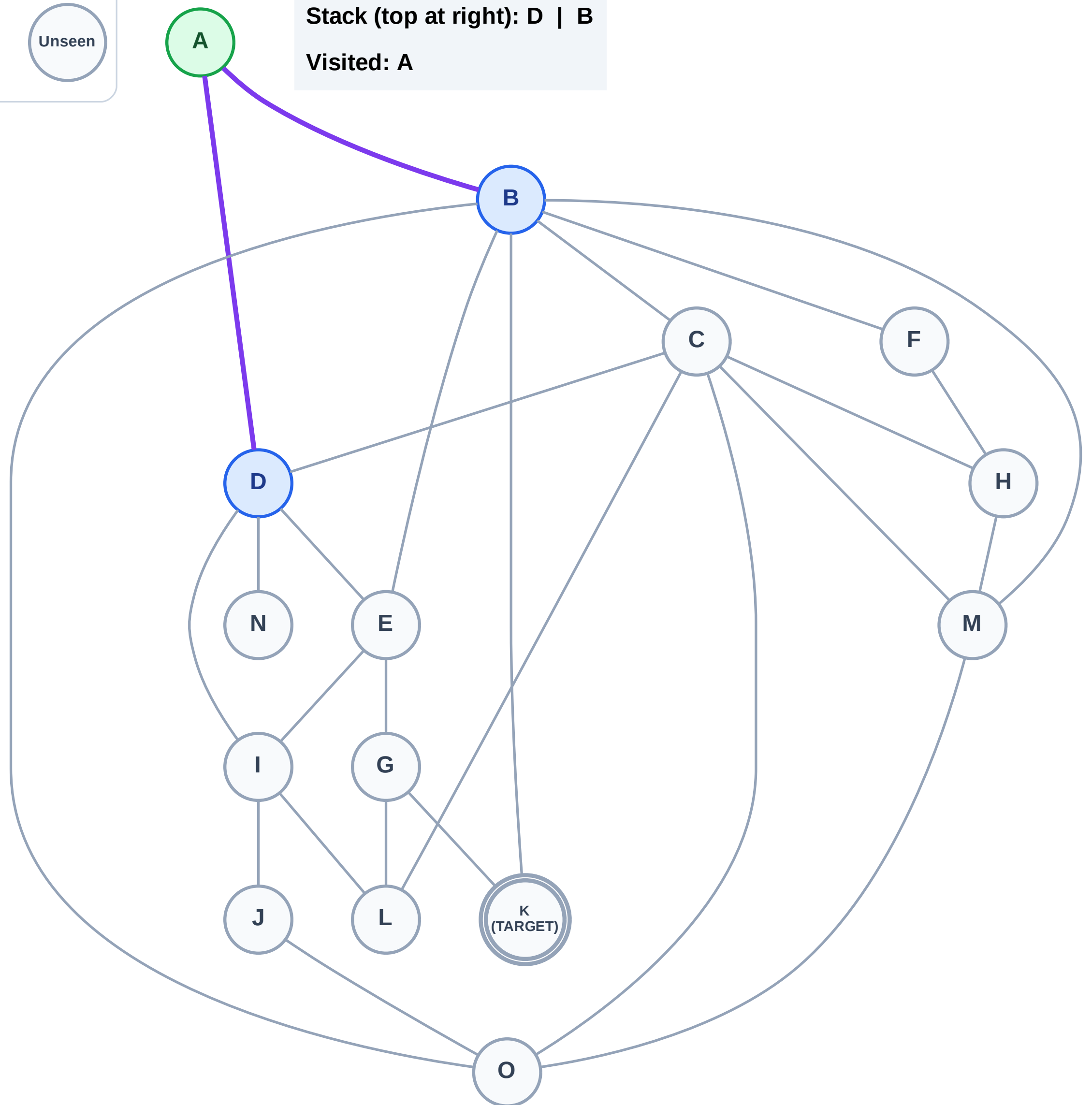
Step 3: Discover D, B from A

Legend



Stack (top at right): D | B

Visited: A



DFS Traversal

Step 4: Process node B

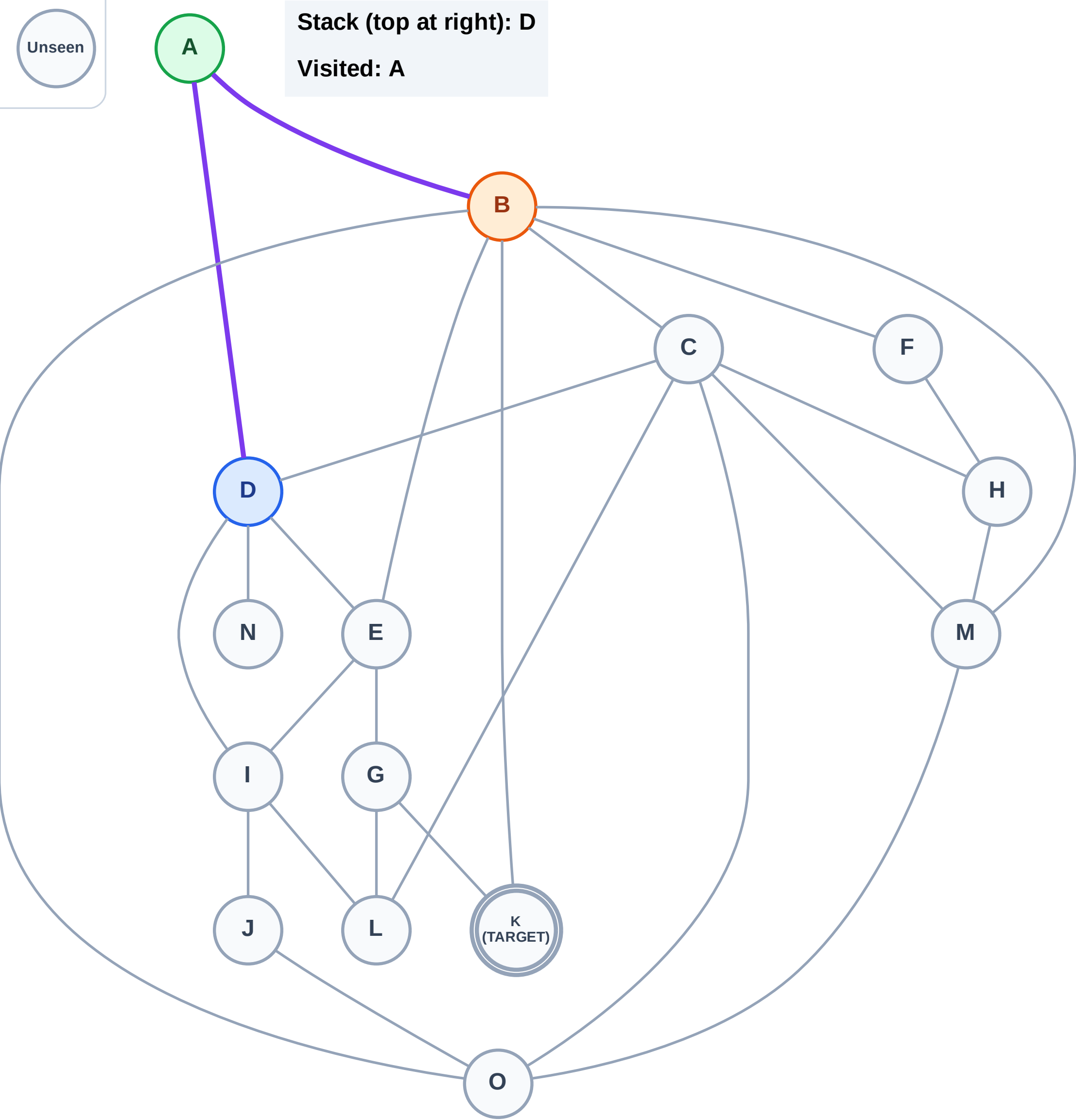
Legend

Complete

Processing

Discovered

Unseen



DFS Traversal

Step 5: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

